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AIMEE International School
Educate & Empower Every Student

Formative Assessment - 1		
Name: _____	Grade: X	Date: _____
Subject: Mathematics	Max Marks: 40	Duration: 1 hr 30 min

Instructions:

1. Read all the questions carefully.
2. Answer the questions in the answer sheets provided. Show your working, wherever necessary.
3. Marks allotted are mentioned in the brackets [ ] at the end of each question or part of the question.
4. Maintain neat handwriting throughout the paper.
5. Submit the answer sheets before leaving the examination hall.
6. Draw neat graphs wherever required.

Section - A **[10 × 1 = 10 M]**

Choose the correct answers:

1. The value of sin 30° + cos 60° is: **[1]**

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{3}$

2. The value of cot 45° is: **[1]**

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) Not defined

3. The value of sec 60° − tan 45° is: **[1]**

(A) 1 (B) 2 (C) $\sqrt{3}$ (D) 0

4. For an acute angle θ, $\sin^2\theta + \cos^2\theta$ equals: **[1]**

(A) 0 (B) 1 (C) 2 (D) $\tan^2\theta$

5. For an acute angle A, $1 + \tan^2 A$ equals: **[1]**

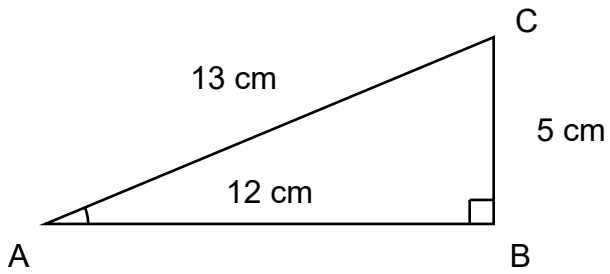
(A) $\text{cosec}^2 A$ (B) $\sec^2 A$ (C) $\cot^2 A$ (D) $\sin^2 A$

6. Which of the following is not defined? **[1]**

(A) tan 0° (B) cos 90° (C) sec 0° (D) cosec 0°

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7. In the figure, triangle ABC is right-angled at B. The value of sin A is: **[1]**



(A) $\frac{12}{13}$ (B) $\frac{5}{13}$ (C) $\frac{5}{12}$ (D) $\frac{13}{5}$

8. If θ is acute and 2 sin θ = 1, then θ equals: **[1]**

(A) 15° (B) 30° (C) 45° (D) 60°

9. If A is acute and cos A = sin 35°, then A equals: **[1]**

(A) 35° (B) 45° (C) 55° (D) 65°

10. The value of $3\tan^2 30° + 2\cos^2 60°$ is: **[1]**

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $\frac{5}{2}$

Section - B **[3 × 1 = 3 M]**

For Questions 11–13, choose the correct option:

(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

11. Assertion (A): For an acute angle A, sin A is less than 1. **[1]**

Reason (R): The hypotenuse of a right triangle is longer than either of its other sides.

12. Assertion (A): tan 60° = 3. **[1]**

Reason (R): For an acute angle θ, tan θ = sin θ divided by cos θ.

13. Assertion (A): tan 45° + cot 45° = 2. **[1]**

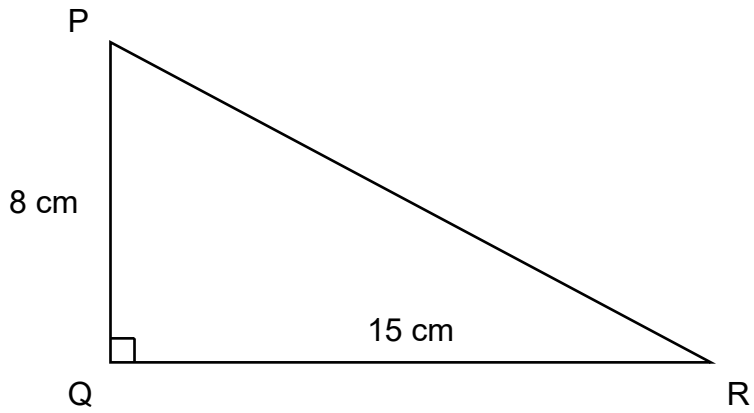
Reason (R): sin 30° = one-half.

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Section - C **[4 × 2 = 8 M]**

This section consists of 4 questions of 2 marks each.

14. In triangle PQR, angle Q = 90°, PQ = 8 cm, QR = 15 cm. Find tan P and sec P. **[2]**



15. Evaluate: **[2]**

$2 \sin 30° \cos 60° + 3 \tan^2 45° - \sec^2 60°$.

16. If A and 2A are acute angles and sin A = cos 2A, find A. **[2]**

17. Simplify, for an acute angle θ: **[2]**

$(1 - \cos^2 \theta) \operatorname{cosec}^2 \theta + \cot^2 \theta - \operatorname{cosec}^2 \theta$.

Section - D **[3 × 3 = 9 M]**

This section consists of 3 questions of 3 marks each.

18. For an acute angle θ, cos θ is given below. Find sin θ, tan θ and sec θ(1 − sin θ). **[3]**

$\cos \theta = \frac{7}{25}$.

19. Prove the following identity for an acute angle θ. Hence, evaluate its left-hand side when θ = 60°. **[3]**

$(\sec \theta - 1)(\sec \theta + 1) = \tan^2 \theta$.

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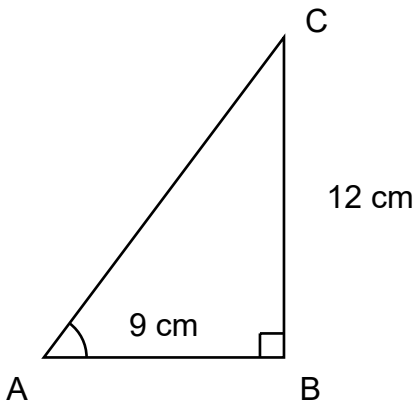
20. Prove the following identity for an acute angle θ: **[3]**

$$\frac{\cos\theta}{1-\sin\theta} - \frac{\cos\theta}{1+\sin\theta} = 2\tan\theta.$$

Section - E **[2 × 5 = 10 M]**

This section consists of 2 questions of 5 marks each.

21. In triangle ABC, angle B = 90°, AB = 9 cm and BC = 12 cm. **[5]**



(a) Find AC. [1]

(b) Find sin A and cos A. [2]

(c) Using these values, verify the following identity. [2]

$(\sin A + \cos A)^2 = 1 + 2\sin A \cos A$.

22. Answer both parts: **[5]**

(a) Evaluate: [2]

$$\frac{\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ}{\tan 45^\circ + \cot 45^\circ}.$$

(b) If θ is acute and 3 tan θ = 4, evaluate: [3]

$$\frac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$$